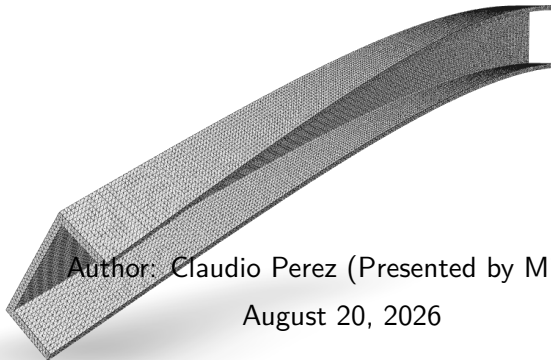


Partial Differential Equations

A Primer for Structural Engineers



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① Introduction

② Questions

③ Anatomy

④ Examples

⑤ Analysis

Partial Differential Equations

- Most mathematical models are derived as simplifications to a PDE.
- Knowing this heritage allows you to use more accurate models when needed.
- Example: Beam theory derived from continuum mechanics.

Objectives

- Appreciate the complexities associated with PDEs
- Learn tools for solving PDEs
- Recognize IVPs, BVPs, and IBVPs
- Get a feel for how understanding PDEs can help you in modeling
- Understand how the curriculum of CE220 and CE225 specialize from CE231

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Note

You generally will not be expected to directly **solve** PDEs yourself. However, being comfortable with how PDEs are *posed*, and being willing to engage with them will help you immeasurably.

① Introduction

② Questions

③ Anatomy

④ Examples

⑤ Analysis

Question: Discretization Error vs Model Error

- **Discretization error:** from approximating a given PDE (grid/mesh/order). Reduced by refinement.
- **Model error:** from choosing an *inadequate PDE*. Refinement does not fix missing terms.
- If the differential equation has been solved, subdividing the mesh won't change anything.

Question: Discretization Error vs Model Error

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Remark

Structural analysis (CE220) is the special case where the underlying differential equation **is** solvable in closed form.

Question: Can I Use Superposition in my finite element analysis?

Question

If I solve my problem for Load Case 1 and Load Case 2 separately, is the solution for both loads acting together simply the sum of the individual solutions?

This is answered by the mathematical property of **linearity**.

Nonlinear PDE

- Contains terms like u^2 , $(u')^2$, or $u \cdot u'$.
- **Superposition fails.**
- Ex: Large Deflections

$$EI \frac{d^4 u}{dx^4} - (EA(u')^2)u'' = f$$

Linear PDE

- The unknown u and its derivatives appear only to the first power.
- **Superposition applies.**
- Ex: Beam Bending

$$EI \frac{d^4 u}{dx^4} = f(x)$$

Question: What is the Model's Fundamental Behavior?

A PDE may be classified by its qualitative behavior. Some common classes are:

- **Elliptic:** Describes steady-state or equilibrium states. The solution at any point is influenced by the data on the *entire* boundary.
 - Ex: Laplace/Poisson's Equation ($\nabla^2 u = f$) for stress analysis.
- **Hyperbolic:** Describes propagation phenomena (waves). Information travels at a finite speed. Characterized by oscillations and requires initial conditions.
 - Ex: The Wave Equation ($\ddot{u} - c^2 \nabla^2 u = 0$) for dynamics.
- **Parabolic:** Describes dissipative processes (diffusion). Tends to smooth out sharp gradients from the initial state over time.
 - Ex: The Heat Equation ($\dot{u} - \alpha \nabla^2 u = 0$)

Implication

Knowing the classification of your governing PDE tells you what kind of behavior to expect, what boundary/initial conditions are required for a well-posed problem, and which class of numerical methods is appropriate.

① Introduction

② Questions

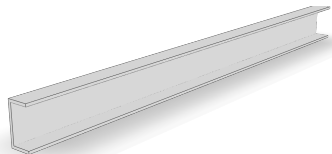
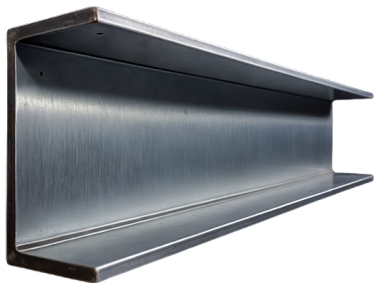
③ Anatomy

④ Examples

⑤ Analysis

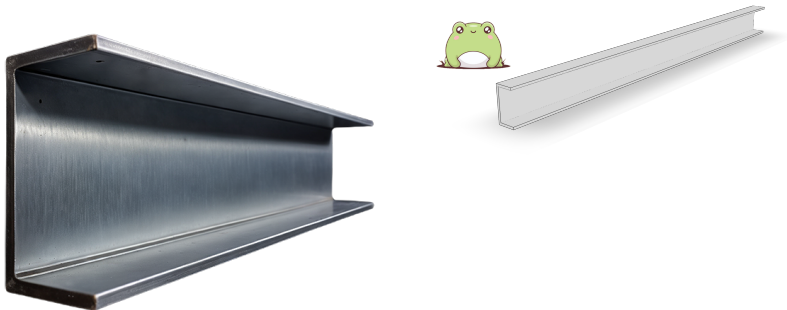
Domains

A *domain* Ω represents the space of *arguments* on which our function u acts.



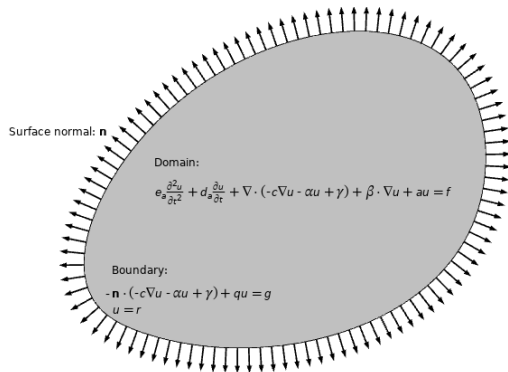
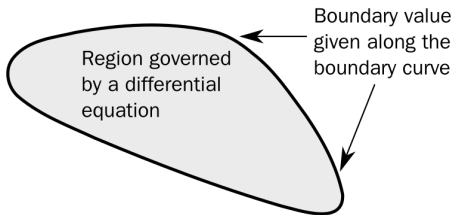
Boundaries

A *boundary* Γ is an edge of a domain... its useful to distinguish this, because its where loads and constraints act.



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(For those of you that learn better with potatoes)

Vectors

Vectors are often introduced as *physical* entities. A more general approach is to define them *algebraically*.

A collection $\{V\}$ of **vectors** with scalars $\{S\}$ is *defined* by the properties:

- **Addition:** Vectors can be combined
 - If $\mathbf{v}$ and $\mathbf{w}$ are vectors in $\{V\}$, then so is $\mathbf{v} + \mathbf{w}$
 - $+$ is *associative* and *commutative*
 - There is a special vector $\mathbf{0}$ in $\{V\}$ with the property: $\mathbf{0} + \mathbf{v} = \mathbf{v} \quad \forall \mathbf{v} \in \{V\}$
- **Scaling:** Vectors can be resized
 - Scaling a vector $\mathbf{v}$ is *associative*: $\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v} \quad \forall \alpha, \beta \in \{S\}$
 - There is a special scalar 1 in $\{S\}$ with the property $1\mathbf{v} = \mathbf{v}$
 - Scaling is *distributive*: $(\alpha + \beta)(\mathbf{v} + \mathbf{w}) = \alpha\mathbf{v} + \beta\mathbf{v} + \alpha\mathbf{w} + \beta\mathbf{w}$
- **Inner Product:** Vectors can be compared
 - Symmetry: $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$.
 - Linearity: $\langle \alpha\mathbf{v} + \beta\mathbf{w}, \mathbf{u} \rangle = \alpha\langle \mathbf{v}, \mathbf{u} \rangle + \beta\langle \mathbf{w}, \mathbf{u} \rangle$ for all $\alpha, \beta \in \{S\}$.
 - Positive-definiteness: $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, with equality iff $\mathbf{v} = \mathbf{0}$.

Functions are vectors

$$\text{index set: } \{1, \dots, n\} \longrightarrow \Omega \subset \mathbb{R}^d, \quad \sum_{i=1}^n \longrightarrow \int_{\Omega}$$

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Finite-dimensional

$$\mathbf{v} = (v_1, \dots, v_n),$$

$$\begin{aligned} \mathbf{v} \cdot \mathbf{w} &= \sum_{i=1}^n v_i w_i \\ &= \langle \mathbf{v}, \mathbf{w} \rangle \end{aligned}$$

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Infinite-dimensional (functions)

$$f(x), \quad \text{“component” of } f \text{ at } x \in \Omega$$

$$\langle f, g \rangle = \int_{\Omega} f(x) g(x) dx \quad (L^2)$$

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Think of the inner product $\langle \cdot, \cdot \rangle$ as a linear function with two arguments, where $\cdot$ is a placeholder for the arguments.

Linear Operators

For a function $f(x_1, \dots, x_n) \rightarrow \mathbb{R}$, ∇ stacks it's partial derivatives:

$$\nabla f = [\partial_{x_1} f, \dots, \partial_{x_n} f]$$

Example

If $f(y, z) = y^2 z$, then

$$\nabla f(y, z) = [2yz, y^2].$$

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- Linearity: for scalars a, b and functions f, g ,

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∇ is just like a matrix. (Except for inverting)

Posing problems

To select a single function from the solution family, we must impose auxiliary conditions.

Boundary Value Problems (BVPs)

Conditions are specified on the physical **boundaries** of the domain Ω . This is the standard formulation for equilibrium or steady-state problems.

Initial Value Problems (IVPs)

Conditions for a time-dependent problem are specified at a single point in time, typically $t = 0$. This provides the "starting state" for the system's evolution.

Most structural dynamics problems are **Initial-Boundary Value Problems (IBVPs)**, requiring both types of conditions.

① Introduction

② Questions

③ Anatomy

④ Examples

⑤ Analysis

Example: Isotropic Elasticity

Find a displacement function $\mathbf{u}(\mathbf{x}, t)$ such that

$$\underbrace{\mu \nabla^2 \mathbf{u}}_{\text{shear part}} + \underbrace{(\lambda + \mu) \nabla(\nabla \cdot \mathbf{u})}_{\text{spherical part}} + \underbrace{\mathbf{f}}_{\text{body force}} = \underbrace{\rho \ddot{\mathbf{u}}}_{\text{inertia}} \quad (\Omega \times (0, T)).$$

where ρ is the density, $\mathbf{f}(\mathbf{x})$ is a function describing distributed loads, and λ and μ are material parameters related to Young's modulus E and Poisson's ratio ν :

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)} \quad \mu = \frac{E}{2(1+\nu)}$$

Example: Continuum Mechanics

Cauchy momentum balance with linear elasticity:

$$\underbrace{\nabla \cdot \boldsymbol{\sigma}}_{\text{stress divergence}} + \underbrace{\boldsymbol{f}}_{\text{body force}} = \underbrace{\rho \ddot{\boldsymbol{u}}}_{\text{inertia}},$$

$$\text{with } \underbrace{\boldsymbol{\sigma}}_{\text{stress}} = \underbrace{\lambda \operatorname{tr}(\boldsymbol{\varepsilon}) \mathbf{1} + 2\mu \boldsymbol{\varepsilon}}_{\text{Hooke's law}},$$

$$\text{and } \underbrace{\boldsymbol{\varepsilon}}_{\text{strain}} = \frac{1}{2}(\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T).$$

This form generalizes readily: replace Hooke's law to model anisotropy, plasticity, etc.

Elasticity (Finite-Deformation)

$$\left\{ \begin{array}{ll}
 \nabla \cdot \underbrace{\left(\lambda \left(\nabla \cdot \mathbf{u} + \frac{1}{2} \|\nabla \mathbf{u}\|^2 \right) (\mathbf{1} + \nabla \mathbf{u}) \right)}_{\text{spherical part}} & \text{in } \Omega \\
 + \underbrace{\mu \left((\nabla \mathbf{u})^T \nabla \mathbf{u} + 2 (\mathbf{1} + \nabla \mathbf{u}) \text{sym } \nabla \mathbf{u} + \nabla \mathbf{u} (\nabla \mathbf{u})^T \nabla \mathbf{u} \right)}_{\text{shear}} + \mathbf{f} = \rho \ddot{\mathbf{u}} & \\
 \mathbf{u} = \mathbf{g} & \text{on } \Gamma_D, \\
 \left((\mathbf{1} + \nabla \mathbf{u}) \left[\frac{\lambda}{2} \text{tr}(\nabla \mathbf{u} + (\nabla \mathbf{u})^T + (\nabla \mathbf{u})^T \nabla \mathbf{u}) \mathbf{1} + \mu (\nabla \mathbf{u} + (\nabla \mathbf{u})^T + (\nabla \mathbf{u})^T \nabla \mathbf{u}) \right] \right) \mathbf{N} = \mathbf{T}_0 & \text{on } \Gamma_N.
 \end{array} \right.$$

Example: Heat Conduction

Temperature $T(\mathbf{x}, t)$:

$$\underbrace{\rho c \dot{T}}_{\text{storage ("charge" rate)}} - \underbrace{\nabla \cdot (\mathbf{q})}_{\text{current divergence}} = \underbrace{r}_{\text{volumetric source}},$$

with $\underbrace{\mathbf{q}}_{\text{heat current}} = - \underbrace{k}_{\text{conductivity}} \nabla T$ (Fourier's law).

Here ρ density, c specific heat, k conductivity (scalar or tensor), r heat generation.

① Introduction

② Questions

③ Anatomy

④ Examples

⑤ Analysis

Weighted Residuals (Weak forms)

Integral theorems: Motivation

- They let us turn **derivative relations** into **integral statements** over a domain/boundary.
- They are the engine behind **weak forms** (finite elements, Galerkin): move derivatives onto test functions.
- They let us **move loads** between boundary and interior (e.g., tractions $\leftrightarrow$ stress divergence).
- All the work comes from two tools:
 - ① a **product rule**, and
 - ② the **divergence/Stokes theorem**.

Gradient and Divergence

Gradient (scalar $\rightarrow$ vector). For $f(x, y, z) = x^2y + z^3$:

$$\nabla f = 2xy \mathbf{i} + x^2 \mathbf{j} + 3z^2 \mathbf{k}$$

Divergence (vector $\rightarrow$ scalar). For $\mathbf{F} = x^2y \mathbf{i} + yz \mathbf{j} + xz^2 \mathbf{k}$:

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = 2xy + z + 2xz$$

Boundary Theorems

Each operator trades a derivative inside a region for values on its boundary.

Gradient theorem (curve, two endpoints):

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a})$$

Divergence theorem (volume, closed surface):

$$\iiint_V (\nabla \cdot \mathbf{F}) dV = \iint_{\partial V} \mathbf{F} \cdot d\mathbf{S}$$

Stokes' theorem (surface, closed curve):

$$\iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_{\partial S} \mathbf{F} \cdot d\mathbf{r}$$

Integral Theorem Cheatsheet

1. Theorem

$$\int_{\Omega} \nabla \cdot \mathbf{v} \, d\mathbf{x} = \int_{\Gamma} \mathbf{v} \cdot \mathbf{n} \, d\mathbf{s}$$

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2. Product rules:

$$\nabla \cdot (\phi \mathbf{w}) = \nabla \phi \cdot \mathbf{w} + \phi \nabla \cdot \mathbf{w}$$

$$\nabla \cdot (\mathbf{A}^T \mathbf{v}) = (\nabla \cdot \mathbf{A}) \cdot \mathbf{v} + \mathbf{A} : \nabla \mathbf{v}$$

$$\nabla \times (\phi \mathbf{w}) = \nabla \phi \times \mathbf{w} + \phi \nabla \times \mathbf{w}$$

Notes: ϕ scalar, $\mathbf{v}, \mathbf{w}$ vectors, $\mathbf{A}$ a linear operator (matrix); $(\mathbf{A} : \nabla \mathbf{v}) = \text{tr}(\mathbf{A}^T \nabla \mathbf{v})$.

Integral theorems: Examples

Recipe: product rule + Gauss (divergence) theorem.

1) Greens first identity.

$$\nabla \cdot (v \nabla u) = \nabla v \cdot \nabla u + v \nabla^2 u \quad \implies \quad \int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Gamma} v \nabla u \cdot n \, dS - \int_{\Omega} (\nabla^2 u) v \, dx.$$

2) Linear momentum (elasticity weak form).

$$\nabla \cdot (\boldsymbol{\sigma}^{\top} \mathbf{v}) = (\nabla \cdot \boldsymbol{\sigma}) \cdot \mathbf{v} + \boldsymbol{\sigma} : \nabla \mathbf{v} \quad \implies \quad \int_{\Omega} \boldsymbol{\sigma} : \nabla \mathbf{v} \, dx = \int_{\Gamma} (\boldsymbol{\sigma} \mathbf{n}) \cdot \mathbf{v} \, dS - \int_{\Omega} (\nabla \cdot \boldsymbol{\sigma}) \cdot \mathbf{v} \, dx.$$

Weighted Residual (Virtual Work)

A weighted residual simply states that the component of the unbalance projected onto an arbitrary vector is zero:

$$\mathcal{P}(\varphi_i) = \langle \varphi_i, (\nabla \cdot \boldsymbol{\sigma} + \mathbf{f}) \rangle = 0 \quad \forall \varphi_i$$

$$p_i(\mathbf{e}_i) = \mathbf{e}_i \cdot (\mathbf{B}\mathbf{q} + \mathbf{p}_w) = 0 \quad \forall \mathbf{e}_i$$

Note that both of these expressions evaluate to real scalars.

Summary

- ODEs/PDEs specify *derivative relations* for an *unknown function*; boundaries specialize to the problem at hand.
- ODE boundaries are numbers; PDE boundaries are functions (infinite-dimensional).

Welcome!